Scarllette Ellis

PhD Researcher within the Bath Reinforcement Learning Lab (she/her)

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A skilled researcher specialising in **reinforcement learning** and **algebraic topology**. My primary research topics are how autonomous agents **learn high-level behaviours** and the **underlying structure** of the **agent-environment interaction**. Experienced in **applying mathematical rigour** to **machine learning research**, and **disseminating scientific research** by **planning** and **delivering** talks, teaching, and events to a range of audiences.

Education — University of Bath

PhD in Reinforcement Learning
Oct 2024 – Oct 2027 (expected)
Supervisors: Professor Özgür Şimşek, Professor Guy McCusker
Proposed Thesis: *The Agent-Environment Topology: A Universal Structure of All Autonomous Learning Agents*
Research areas: Graph Theory, Information Theory, Topology, and Hierarchical Skill Discovery for Reinforcement Learning,

MRes in Accountable, Responsible, and Transparent AI
Oct 2023 – Oct 2024
Grade: Distinction
Dissertation: *Unifying Graph and Information Skill Discovery in Reinforcement Learning*
Supervisors: Professor Özgür Şimşek, Professor Guy McCusker

Masters in Mathematics and Computer Science
Sep 2019 – July 2023
Grade: First Class (Hons)
Masters Dissertation: *Getting Prepared: Finding Subgoals by using the Graphical Structure of Environments to Maximise Empowerment*
Undergraduate Dissertation: *Reinforcement Learning in the Board Game Pandemic*
Supervisor: *Dr. Joshua B. Evans*
My degree focused on linear algebra, group theory, algebraic topology, parallel programming, and machine learning.

Teaching and Outreach Work

Dissertation Supervisor - *University of Bath*
Sep 2024 – Present
I have supervised 14 undergraduate research projects on areas of reinforcement learning, including implementing state-of-the-art reinforcement learning in tabletop games and deploying reinforcement learning in applications like resource trading and dynamic difficulty adjustment. When supervising, I help students identify meaningful research questions, how to decompose questions into clear steps, and how to present findings with precise writing and visualisations.

Teaching Assistant - *University of Bath*
Sep 2022 – Present
Modules: Artificial Intelligence, Reinforcement Learning, Bayesian Machine Learning.
I have taught undergraduate and postgraduate students python programming, scientific writing, and mathematics for machine learning. Providing guidance to students on technical work during busy tutoring sessions. In these roles, I have been responsible for marking student work and providing actionable feedback, including fixing and deploying automarking software for machine learning coursework.

Maths Support Coach - *Future Foundation, Bootham School York*
Sep 2026
A residential summer school where I taught students aged 11–13. I delivered engaging and exciting lessons on maths and its applications, while ensuring student safety and wellbeing. I applied my science communication skills to plan and run a Dungeons & Dragons games for students, encouraging them to develop their teamwork, communication, and creative problem-solving skills.

Skills

Python programming utilising Jax and NumPy. Applying these to build deep, intrinsically motivated, and hierarchical reinforcement learning algorithms.

Creating a range of reinforcement learning environments that evaluate specific agent behaviour, making use of a generic agent-environment framework.

Using Matplotlib and Inkscape to designing clear and precise scientific diagrams, including visualising agent behaviour and complex graph networks.

Using LaTeX and Manim to create and deliver engaging talks on maths and reinforcement learning.

Experience

Automarking Internship - *University of Bath*
June 2023 – Sep 2023
I investigated the automarking tools used throughout the computer science department. Conducting interviews with lecturers and students. My investigation now informs how automarking software is selected and deployed in the department; whether it can be relied upon to save time and provide clear feedback, and how marking errors can be avoided

Talks and Events

Whose Science Is It Anyway? Using Improv theatre and Dungeons & Dragons to enhance science communication skills
July 2025
BIG Event - Royal Institution London

Being Prepared: Creating Multi-Level Skill Hierarchies using Graph and Information Theory
Oct 2024
University of Bath Centre for Mathematics and Algorithms for Data

The Current State and Best Practices for University Automarking
Sep 2023
University of Bath Lecturer Training Day